

3.10.2 Physics of the ear (A-level only)

3.10.2.1 Ear as a sound detection system (A-level only)

Content

Simple structure of the ear, transmission processes.

3.10.2.2 Sensitivity and frequency response (A-level only)

Content

Production and interpretation of equal loudness curves.

Human perception of relative intensity levels and the need for a logarithmic scale to reflect this.

Definition of intensity.

Intensity level = $10 \log \frac{I}{I_0}$ where the threshold of hearing $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$

Measurement of sound intensity levels and the use of dB and dBA scales; relative intensity levels of sounds.

3.10.2.3 Defects of hearing (A-level only)

Content

The effect on equal loudness curves and the changes experienced in terms of hearing loss due to injury resulting from exposure to excessive noise or deterioration with age (excluding physiological changes).